LightMamba: A Resource-Efficient Deep-Learning Method for UWB NLOS Identification Based on Selective State-Space Modeling

Tao Wang, Yuanming Shu, Lulu Qiao, Dong Ding, and Guangxue Li

***Abstract*—Ultrawideband (UWB) technology is pivotal for centimeter-level indoor positioning, yet its accuracy is severely degraded by non-line-of-sight (NLOS) propagation in complex industrial environments. Existing NLOS identification methods face critical trade-offs between model accuracy, computational efficiency, and dynamic scenario adaptability. To address these challenges, we propose a novel lightweight framework named LightMamba that integrates selective state-space models (SSMs) with UWB channel impulse response (CIR) characteristics. LightMamba introduces three key innovations: 1) a dual-branch feature extraction module that combines segmented CIR projection with auxiliary channel feature alignment, preserving both local temporal dependencies and global environmental semantics; 2) a Mamba block with dynamic path selection, enabling adaptive focus on NLOS-induced multipath components while maintaining linear computational complexity and efficient processing of long sequences; 3) a resource-efficient architecture achieving millisecond-level inference latency without sacrificing accuracy. Extensive experiments on three real-world datasets (eWINE, IULCD, OIUD) demonstrate LightMamba's superiority, achieving a high *F*1-score of 95.20% on the OIUD dataset with only 22k parameters, outperforming baseline models while reducing the parameter number by 96% compared to long short-term memory (LSTM) with 545k parameters. This work provides a new solution for high-reliability UWB-based indoor positioning systems in industrial Internet of Things (IIoTs) applications.**

***Index Terms*—Industrial Internet of Things (IIoTs), LightMamba, non-line-of-sight (NLOS), selective state-space model (SSM), ultrawideband (UWB).**

I. Introduction

FROM consumer-grade indoor navigation to mission-critical industrial automation, the proliferation of location-aware services has driven an urgent demand for centimeter-level positioning accuracy in indoor environments. While the global navigation satellite system (GNSS) achieves meter-level to centimeter-level accuracy outdoors, its signals suffer over 30 dB attenuation in indoor scenarios such as factories and offices, rendering GNSS ineffective for real-time high-precision indoor positioning [1]. To overcome this bottleneck, alternative solutions based on wireless fidelity (WiFi) [2], Bluetooth low energy (BLE) [3], and ultrawideband (UWB) have been proposed [4]. Among these, UWB technology, leveraging nanosecond-level pulse signals combined with time-of-flight (ToF) measurements, achieves optimal positioning accuracy of 15 cm, demonstrating significant technical advantages. Unlike WiFi/BLE systems that are inherently susceptible to multipath interference, UWB's gigahertz-level bandwidth enables precise resolution of microscopic time-delay characteristics in channel impulse response (CIR) [5]. This makes UWB the preferred solution for high-reliability positioning in industrial Internet of Things (IIoTs) scenarios, including collaborative obstacle avoidance of autonomous mobile robots (AMRs) and augmented reality (AR)-assisted equipment maintenance [6].

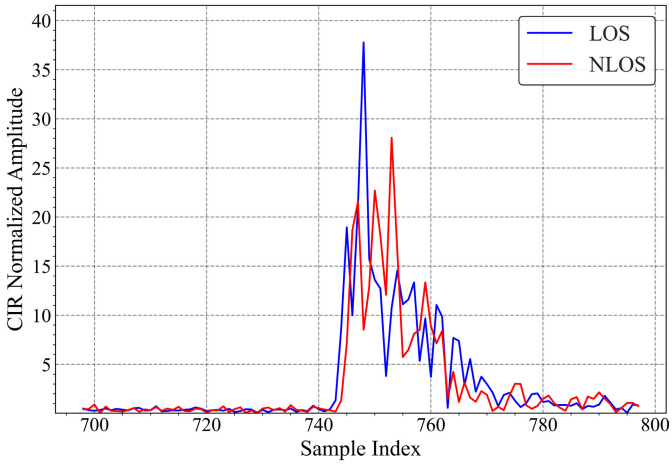


Fig. 1. LOS and NLOS CIR samples of UWB signals.

However, UWB deployment in complex industrial environments faces two main challenges as follows.

1) *NLOS:* Line-of-sight (LOS) refers to the scenario where there is an unobstructed path between the UWB transmitter and receiver. In contrast, non-line-of-sight (NLOS) propagation is induced by dense obstructions and dynamic human-machine interactions, significantly increasing positioning errors [7].
2) Fig. 1 illustrates the LOS and NLOS CIR samples of UWB signals. This poses severe risks to high-value

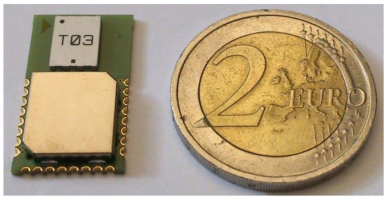
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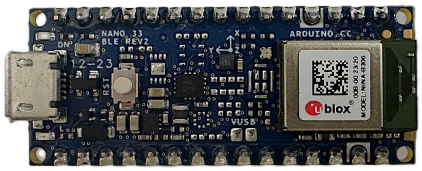
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(a)



(b)

Fig. 2. (a) Decawave DWM1000 module [11]. (b) Arduino Nano nRF52840 MCU used in our analysis of resource consumption.

assets like semiconductor-transporting automated guided vehicles (AGVs) [8].

3) *Algorithm-Hardware-Scenario Mismatch:* Current NLOS identification schemes exhibit critical limitations as follows.
   a) *Accuracy-Oriented Methods:* Deep neural networks (e.g., 1D-CLANet, FCN-Attention) achieve > 90% recall [9], [10], but their million-parameter models exceed the 256 KB memory capacity of typical UWB edge nodes (e.g., Decawave DWM1000 [11] shown in Fig. 2).
   b) *Efficiency-Oriented Methods:* Tiny machine-learning (TinyML) techniques compress models to ∼10 KB while maintaining performance through manual feature engineering [12]. Nevertheless, the generalization capability of methods like TinyML will be limited in complex industrial environments due to their reliance on features extracted from predetermined, static segments of the CIR. This approach of manual feature engineering, by focusing on fixed CIR windows, risks overlooking crucial NLOS characteristics that may manifest outside these segments in diverse or extreme scenarios, thereby hindering the model's adaptability to varying channel conditions.
   c) *Inherent Architectural Mismatches With UWB Signal Characteristics:* Standard deep-learning architectures often face fundamental conflicts when applied to UWB NLOS identification due to inherent mismatches with UWB signal properties. Recurrent neural networks' (RNNs) fixed scanning order cannot capture noncausal multipath features in NLOS-distorted CIRs; transformers' and other attention mechanisms' quadratic complexity conflicts with edge energy constraints; convolutional neural networks' (CNNs) isotropic kernels struggle to model directional correlations in metallic reflections. These limitations underscore the need for an alternative modeling framework.

Emerging selective state-space models (SSMs), exemplified by Mamba [13], offer a novel solution to achieve high performance while maintaining efficient inference, which is crucial for practical applications. SSMs theoretically support sparse long-range modeling of CIRs through dynamic path selection and linear complexity. Their hardware-friendly matrix computation and physics-informed parameterization unlock new potential for industrial NLOS identification. This algorithm-physics-hardware co-design philosophy underpins the core idea of our LightMamba framework: achieving high-reliability edge inference with millisecond-level latency through mechanism-driven reconfiguration of SSM for UWB signals.

In this article, we propose a lightweight UWB NLOS identification framework for industrial edge devices, the LightMamba, whose core contributions are reflected in the following three aspects.

1) *First Application of Mamba to UWB NLOS Identification:* This article pioneers the use of the selective SSM of Mamba for UWB NLOS identification, validating its effectiveness and potential.
2) *LightMamba Framework:* By strategically reconfiguring SSMs based on UWB signal mechanisms, this article introduces a LightMamba framework that effectively balances high performance with efficient inference for UWB signal processing.
3) *Real-World Datasets Validation:* Experiments on publicly available datasets from authentic industrial environments demonstrate LightMamba's robustness and efficiency in complex NLOS scenarios.

LightMamba provides a technical pathway for lightweight temporal models in UWB signal processing. Its performance-efficiency-balanced design advances the practical deployment of industrial IoT positioning systems.

II. Related Works

A. NLOS Identification Based on Traditional Signal Features

Prior to the advent of deep learning, NLOS identification predominantly relied on handcrafted feature extraction combined with classical machine learning algorithms. Early approaches analyzed statistical characteristics of the CIR, such as kurtosis, skewness, and maximum amplitude decay rate, to construct classifiers like support vector machines (SVMs) or random forests. For instance, Musa et al. [14] proposed a recursive decision tree method for UWB-based NLOS identification, achieving 90% accuracy in distinguishing LOS/NLOS scenarios. Similarly, Wang et al. [15] enhanced NLOS classification using an improved snake search algorithm, surpassing 90% in accuracy, recall, and other metrics. Fan and Awan [16] demonstrated 86.5% accuracy with unsupervised machine learning, while Zhu et al. [17] applied modified fuzzy decision tree to achieve over 90% accuracy in both dynamic and static indoor environments. Yang et al. [18] further improved localization accuracy by 63.4% using SVM with a recall rate exceeding 92% for LOS/NLOS classification. Despite their low computational complexity, these methods suffer from dependency on manual feature engineering, limiting their generalization in industrial environments with dynamic metallic reflections and noises.

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B. Deep Learning-Driven NLOS Identification

Deep learning has enabled end-to-end learning from raw signals to classification outcomes, significantly advancing NLOS identification. CNN, long short-term memory (LSTM) networks, attention mechanisms, and hybrid architectures have demonstrated superior performance in extracting spatiotemporal features for LOS/NLOS discrimination. Bregar and Mohorčič [19] pioneered the use of CNNs for raw CIR-based NLOS classification, revealing the potential of deep learning in this domain. Poulose and Han [20] optimized UWB positioning with LSTM-based NLOS classification, achieving an average localization error of 7 cm. Tran et al. [21] leveraged a generative adversarial network (GAN)-based model combined with optimization techniques to attain 86.98% accuracy, whereas Wei et al. [22] reduced localization errors to 7.35 cm using a GRU-CNN hybrid and multiple instance learning (MIL). Recent advancements include Kim et al. [23] achieving 82% accuracy with LSTM models, Wang et al. [9] proposing the 1D-CLANet architecture achieving an accuracy of 95.58%, and Pei et al. [10] developing a self-attention-enhanced fully convolutional network attention (FCN-Attention) that achieved 88% accuracy in benchmark datasets and 100% accuracy in domain-specific datasets. However, these models demand substantial computational resources. For example, FCN-Attention architectures need more than one million parameters, resulting in prolonged inference times that hinder deployment on resource-constrained devices.

C. NLOS Identification on Edge Devices

To address the computing burden with a resource-constrained deployment, researchers have explored model compression techniques such as pruning, quantization, and TinyML frameworks. Barral et al. [24] integrated received signal strength (RSS) with SVM to achieve high NLOS classification success rates on low-cost UWB devices. Fontaine et al. [25] enhanced edge inference accuracy by 26% using semi-supervised encoding, reducing localization errors to 5.9 cm with a 35 000-parameter model. Dong introduced a Fresnel zone-based method with minimal prior knowledge, achieving a 6.45-mm localization accuracy in simulations [26]. Wu et al. combined quantization, feature selection, and TinyML to enable efficient NLOS identification on edge devices with state-of-the-art (SOTA) [12]. While these methods reduce memory footprints by $2\times$ with acceptable accuracy loss, their reliance on prior feature selection risks discarding critical spatiotemporal information from UWB signals. This information loss complicates multisensor fusion, e.g., inertial measurement unit (IMU)/UWB collaboration, necessitating data alignment through downsampling or interpolation, which introduces additional computational cost and error propagation.

D. Robust Algorithms to Mitigate NLOS Effects

Enhancing the positioning reliability is another critical research field. Many algorithms of multisensor fusion and robust filtering have been proposed to mitigate the impact of NLOS-induced errors and other data anomalies. Liu et al. [27] achieved this in pedestrian dead reckoning by tightly integrating smartphone sensors and applying motion constraints. Robust filtering algorithms like the robust Kalman filter (RKF) [28] and the robust cubature Kalman filter (RCKF) [29] were employed to restrain the influence of low-quality observations, sensor noise and NLOS errors for integrated navigation. While effective, these algorithms often rely on complex system modeling, highlighting the need for a more lightweight and end-to-end solution.

E. Mamba-Based Selective SSM: Revolutionizing Long-Sequence Signal Processing

The Mamba, a selective SSM proposed in 2023, offers a new option for balancing computational efficiency and long-sequence modeling [13]. Unlike attention-based transformers, Mamba achieves linear time complexity via hardware-optimized state transition mechanisms [30]. Its dynamic parameterization enables context-aware filtering of signal components, exhibiting particular advantages in industrial environments with transient metallic reflections. Preliminary applications of Mamba in medical image analysis [31], language modeling [32], and large-scale UWB-based localization [33] have demonstrated its capability to model temporal dynamics and contextual information while maintaining computational efficiency. This innovative model is very promising for real-time NLOS identification and error mitigation in dynamic IoT scenarios.

III. Theoretical Analysis

A. Theoretical Foundation for NLOS Signal Processing

A solid theoretical framework is essential for understanding and modeling the complexities of UWB signals under NLOS conditions. From an algorithmic perspective, UWB NLOS signals present two primary challenges to traditional sequential models: the noncausal characteristic, and the sparse distribution of critical features.

The first challenge, the noncausal characteristic, arises from the fundamental difficulty in identifying the true direct path (DP). In NLOS scenarios, the DP, representing the shortest physical propagation route, often experiences severe attenuation and may become indistinguishable from noise. Consequently, the first detectable path (FDP) that an algorithm identifies is frequently a later, stronger multipath component, not the true DP. This misidentification introduces systematic biases into time of arrival (ToA) estimation [34]. This phenomenon fundamentally disrupts the simple causal assumption inherent in many sequential models, where the first significant event in a sequence is assumed to be the primary physical onset. Models that rely on such strict temporal causality struggle to differentiate between the apparent onset of the FDP and the physical onset of the DP, leading to poor identification performance.

The second challenge is the sparse distribution of features. The signal distortions that serve as key indicators of NLOS conditions, such as those caused by multipath and diffraction, are not confined to a specific region of the CIR. Instead, these critical features are often sparsely scattered throughout

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the entire long-sequence CIR. This sparsity demands that a model possesses a strong capability for capturing long-range dependencies, enabling it to discover and retain the significance of these sparse yet vital features from a sequence composed mostly of noncritical information.

To effectively address the above two challenges, an ideal model must move beyond simple causal assumptions and be adept at capturing sparse, long-range dependencies with high computational efficiency.

B. Mamba: a Selective SSM for NLOS Identification

Based on the requirements theoretically outlined above, we select the Mamba architecture, which is founded on a selective SSM. We believe that the mathematical mechanisms of Mamba are particularly well-suited to the demands of NLOS signal processing.

The core of Mamba is its ability to compress a long history of sequential information into a compact hidden state, $h(t)$, which evolves over time. This dynamic process is governed by the following continuous state-space equations for an input sequence $x(t)$:

$$\frac{dh(t)}{dt} = \boldsymbol{A}h(t) + \boldsymbol{B}x(t) \tag{1}$$

$$y(t) = \boldsymbol{C}h(t) \tag{2}$$

where $h(t) \in \mathbb{R}^N$ is the hidden state, $x(t) \in \mathbb{R}$ is the input, and $y(t) \in \mathbb{R}$ is the output. The matrices $\boldsymbol{A} \in \mathbb{R}^{(N\times N)}$, $\boldsymbol{B} \in \mathbb{R}^{(N\times 1)}$, and $\boldsymbol{C} \in \mathbb{R}^{(1\times N)}$ are the state parameters that govern the system's dynamics. A crucial innovation in Mamba is that these parameters, particularly $\boldsymbol{B}$ and $\boldsymbol{C}$, are not static but are dynamically parameterized based on the input data $x(t)$. This input-dependent selectivity is crucial to its effectiveness.

For computational processing, the continuous hidden state updates can be discretized as

$$h_k = \bar{\boldsymbol{A}}h_{k-1} + \bar{\boldsymbol{B}}x_k \tag{3}$$

$$y_k = \boldsymbol{C}h_k \tag{4}$$

where $\bar{\boldsymbol{A}}$ and $\bar{\boldsymbol{B}}$ are the discretized versions of $\boldsymbol{A}$ and $\boldsymbol{B}$, dependent on the discretization timestep $\boldsymbol{\Delta}$. The state matrix $\bar{\boldsymbol{A}}$ acts as a memory retention gate. By parameterizing $\bar{\boldsymbol{A}}$ appropriately, the model can carry the memory of a critical feature across long, uneventful segments of the sequence with linear complexity of $O(L)$. When a new input x_k is encountered, the selective matrix $\bar{\boldsymbol{B}}$ can immediately update the state by taking fully into account the sparse but critical features. This selective mechanism allows Mamba to maintain and update sparse but critical information throughout the whole input sequence.

C. Mechanistic Advantages of Mamba Over the Transformer Architecture

While the transformer architecture also excels at modeling long-range dependencies, we argue that Mamba's architecture offers more fundamental, mechanistic advantages for the specific problem of UWB NLOS identification.

A primary limitation of the transformer is the potential attention dilution of its self-attention mechanism. When processing a long CIR sequence, the global weighted-averaging process of self-attention can cause the significance of a few sparse, critical features to be diminished by the overwhelming amount of noncritical information or noise. In contrast, Mamba's selective state compression fundamentally avoids this problem. By selectively updating its compact state based on the current input rather than performing a global average, it preserves the sparse features with much greater fidelity and efficiency.

Furthermore, while the transformer can model the relationship between any two points in the sequence to address the noncausal characteristic, it does so at a high computational cost of $O(L^2)$. Mamba, through its inherent sequential processing as expressed by (3), learns to recognize the dynamic evolution of the signal pattern, such as the characteristic NLOS signature being weak earlier and strong later. This strategy of learning the overall temporal pattern is not only more computationally efficient but also more aligned with the nature of signal processing tasks.

In summary, Mamba's advantages lie beyond its linear complexity for this particular task. Its architecture is mechanistically better aligned with the physical and algorithmic challenges posed by UWB NLOS signals, making it a more suitable and efficient foundation for our model design.

IV. Model Design

This article introduces LightMamba, a novel lightweight framework for NLOS identification of long-sequence UWB signals. This framework aims to effectively use the core characteristics of CIR data, specifically the multipath time dispersion resulting from the distribution of effective features within the first few hundred sampling points. Additionally, the model considers auxiliary features, such as the signal kurtosis, and their nonlinear relationship with propagation conditions. It also addresses the issue of inconsistent CIR data lengths due to different sampling strategies across various UWB devices. As illustrated in Fig. 3, the LightMamba model architecture primarily consists of the following five key modules: a dual-branch feature extraction module for processing CIR encoding and auxiliary features separately, a feature fusion module to combine the extracted features, the core sequence modeling module Mamba block, a middle block, and a classifier. The processing methodologies and dimensional transformations within each module are also detailed in Fig. 3. The core innovation of LightMamba lies in the introduction of the selective mechanism in the Mamba model to overcome the limitations of static input for traditional SSMs and their performance limitations when dealing with complex and dynamically changing signals. Sections IV-A–IV-D will elaborate on each of these modules.

A. Dual-Branch Feature Extraction Module

The dual-branch feature extraction module (Feature Extractor) serves as the input processing front end of the LightMamba framework, comprising two parallel branches: the CIR feature extraction branch and the auxiliary feature extraction branch for separately processing different input data.

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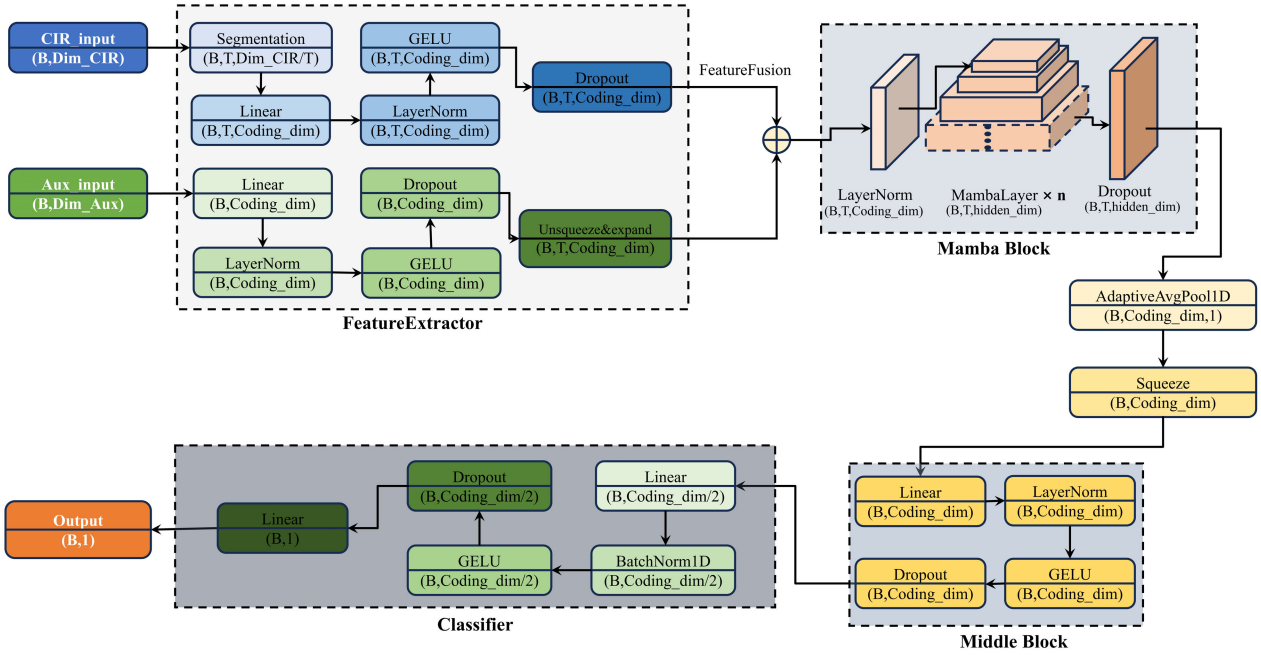


Fig. 3. Architecture of the proposed LightMamba model.

1) CIR Feature Extraction Branch: For the original CIR input data $\boldsymbol{X}_{\mathbf{CIR}} \in \mathbb{R}^{\boldsymbol{B}\times \boldsymbol{D_c}}$, where $\boldsymbol{B}$ denotes the batch size and $\boldsymbol{D_c}$ represents the length of the CIR data, this article employs a segmented projection strategy to efficiently extract local temporal features and reduce computational complexity. This strategy involves the following two steps: time segmentation and latent space mapping.

First, the original CIR data, which contains rich information about signal propagation paths, is divided along the time dimension into $\boldsymbol{T}$ consecutive subsegments. This step aims to capture local temporal dependencies. For instance, when $\boldsymbol{D_c} = 1016$, if $\boldsymbol{T}$ is set to 8, each subsegment contains 127 time points. It is noteworthy that by appropriately selecting the number of subsegments $\boldsymbol{T}$, the global representation of the information contained in the CIR data can be preserved while the efficiency of subsequent processing can be improved. The mathematical expression for time segmentation is as follows:

$$X_{\text{seg}} = \text{Reshape}\left(X_{\text{CIR}}, \left(B, T, \left[\frac{D_c}{T}\right]\right)\right). \tag{5}$$

This operation effectively retains the local temporal structure, preventing the potential blurring of multipath features caused by global pooling. It is important to note that the number of segments $\boldsymbol{T}$ must be chosen such that it evenly divides the original CIR length $\boldsymbol{D_c}$ to ensure no data points are discarded during the reshaping process. This constraint requires the selection of $\boldsymbol{T}$ to be based on the CIR dimensionality of the specific dataset.

Following time segmentation, the CIR extraction branch uses a linear layer to map the data of each local time segment into a low-dimensional latent space. This step aims to extract a compact yet informative representation from each segment, capturing local temporal patterns and multipath characteristics that are discriminative for LOS/NLOS classification. The mathematical expression for this process is

$$H_{\text{CIR}}(t) = \text{GELU}\left(\text{LayerNorm}\left(W^c X_{\text{seg}}(t)\right)\right) \tag{6}$$

where $\boldsymbol{X_{\text{seg}}(t)} \in \boldsymbol{R}^{\boldsymbol{B}\times[\boldsymbol{D_c}/\boldsymbol{T}]}$ represents the data of tth time subsegment, $W^c \in \boldsymbol{R}^{[\boldsymbol{D_c}/\boldsymbol{T}]\times \boldsymbol{d}}$ is a learnable linear transformation matrix, d is the dimension of the mapped space, LayerNorm denotes the layer normalization operation, the GELU is the Gaussian error linear unit activation function used to enhance the model's nonlinear representation capability.

2) Auxiliary Feature Extraction Branch: For the auxiliary feature input $\boldsymbol{X_{\text{aux}}} \in \mathbf{R}^{\boldsymbol{B}\times \boldsymbol{D_a}}$, where $\boldsymbol{D_a}$ represents the dimension of the auxiliary features, a cross-model alignment strategy is adopted for effective fusion with CIR features. This processing includes the following two steps.

Dimension Compression: A linear layer is used to project the $\boldsymbol{D_a}$-dimensional auxiliary features into the same $\boldsymbol{d}$-dimensional space as CIR features. The mathematical expression is

$$H_{\text{aux}}(t) = \text{GELU}\left(\text{LayerNorm}\left(W^a X_{\text{aux}}\right)\right) \tag{7}$$

with W^a being the weight matrix.

Temporal Expansion: To ensure that the environmental semantic information contained in the auxiliary features covers all CIR time segments, simulating the uniform impact of environmental factors such as obstacles on the entire time-domain signal, the auxiliary features are repeated along consecutive time segments. The mathematical expression is

$$\hat{H}_{\text{aux}} = \text{Repeat}\left(H_{\text{aux}}, [1, T, 1]\right). \tag{8}$$

Through the operations described above, the model successfully maps CIR data and auxiliary features of different dimensions into the same feature space, laying the groundwork for subsequent feature fusion.

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B. Feature Fusion

To effectively combine the temporal information from CIR and the environmental semantic information from the auxiliary features, this article employs a feature fusion operation by element-wise addition, rather than simple concatenation. After the processing in the previous steps, the CIR features and the expanded auxiliary features have the same dimensions. For the aligned features, fusion is performed using element-wise addition, which preserves the original temporal structure of CIR while uniformly injecting the global environmental information contained in the auxiliary features. The mathematical expression is as follows:

$$H_{\text{fused}} = H_{\text{CIR}} \oplus \hat{H}_{\text{aux}} \tag{9}$$

where $\oplus$ denotes the element-wise addition operation.

C. Mamba-Based Sequence Modeling

The Mamba module is the core component of the LightMamba framework. By introducing a selective scan mechanism, it enables the model to dynamically adjust the time-step parameter Δ based on the input, thereby capable of adaptively focusing on critical time intervals. For UWB signals, especially in NLOS scenarios where sudden increases in multipath components may occur, Mamba can autonomously adjust the size of Δ to perform fine-grained analysis of these local features. To further enhance the model stability during training, LightMamba incorporates a residual connection and a layer normalization before the Mamba modeling, with the following mathematical expression:

$$H_{\text{out}} = H_{\text{in}} + \text{Dropout}\left(\text{Mamba}\left(\text{LayerNorm}\left(H_{\text{in}}\right)\right)\right) \tag{10}$$

where $\boldsymbol{H}_{\text{in}}$ is the input to the Mamba module and $\boldsymbol{H}_{\text{out}}$ is the output. **Dropout** is used to prevent overfitting. It should be noted that our analysis and experimental results indicate that unlike transformer models based on attention mechanism, the performance improvement of Mamba model does not significantly increase with simple stacking of layers. This is primarily because the update and transfer of the hidden state in the linear space of Mamba have a relatively low correlation with the number of layers, leading to a noticeable diminishing return effect. Excessive stacking of Mamba layers provides limited performance gains and instead significantly increases the model size and number of parameters, which aligns with our experimental observations.

D. Classification Head Design

To map the features obtained from sequence modeling to final classification results, this article designs a classification head that mainly includes adaptive temporal pooling, an intermediate mapping layer, and a batch normalization (BN) layer. Traditional average pooling treats noise-dominated time segments equally, potentially degrading classification performance. To avoid this disadvantage, this article adopts adaptive temporal pooling, allowing the model to autonomously down-weight noise-dominated time segments. To further prevent overfitting, an intermediate mapping layer containing Dropout and GELU activation function is introduced after pooling layers. Finally, a BN layer and a linear layer are used for the final classification. The BN layer is used to mitigate feature distribution shifts between different batches of data, improving the model's generalization ability.

V. Experiment

A. Data Description and Preprocessing

To evaluate the performance and generalization capability of LightMamba across different datasets, this article selected three publicly available datasets, including the eWINE dataset [35], the Industrial UWB Localization CIR Dataset (IULCD) [25], [36], and the Office Industrial University Dataset (OIUD) [37] provided by Ghent University. The eWINE dataset serves as a standard benchmark, comprising 42 000 samples from seven distinct scenarios, such as offices and workshops. Each sample provides a 1016-point CIR and 14 auxiliary features. The IULCD and OIUD datasets were used to assess the model's performance in realistic industrial environments. The IULCD provides 20 360 samples, each with a 1016-point CIR, from an industrial environment as shown in Fig. 4. It should be noted that the NLOS labels for the IULCD were not provided directly; we manually added them for the 13 usable data files by strictly following the rules in the dataset's README document. The OIUD offers 65 979 samples from industrial and office environments, while its university subset was excluded due to extensive data loss and missing labels. All three datasets were collected using Decawave DWM1000 devices.

A critical data preprocessing step was performed for the IULCD and OIUD datasets. These datasets originally contained features like estimated range and actual range that act as direct proxies for the NLOS ground truth. To ensure the model learns from signal characteristics rather than leaked range information, these and other potentially leaking variables were meticulously removed. This rigorous preprocessing resulted in a final auxiliary feature count of 7 for the IULCD and OIUD datasets.

For training and evaluation purposes, each dataset was split into training, validation, and test sets with a ratio of 0.7, 0.2 and 0.1, respectively.

B. Training Details

To accommodate the varying attributions of different datasets, the input dimensions were configured depending on the dataset while maintaining the same model structure and hyperparameter configuration. Given the varying CIR lengths $\boldsymbol{D_c}$ across different datasets (150 for OIUD, 1016 for eWINE, and IULCD) and the requirement that the number of time segments $\boldsymbol{T}$ must evenly divide $\boldsymbol{D_c}$ as explained in Section IV, we selected $\boldsymbol{T} = 5$ for the OIUD dataset and $\boldsymbol{T} = 8$ for the eWINE and IULCD datasets. The input dimensions for each dataset are summarized in Table I.

To further enhance the training stability and performance of the model, the following strategies are also employed during training: gradient clipping, conservative initialization, NaN value monitoring, class-weighted loss function, learning rate scheduling, and early stopping approach. The model training is

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TABLE I
INPUT DIMENSIONS FOR THE THREE DATASETS

Attribution term	OIUD	eWINE	IULCD
CIR Length	150	1016	1016
Number of Aux. Features	7	14	7
Number of Time Segments	5	8	8

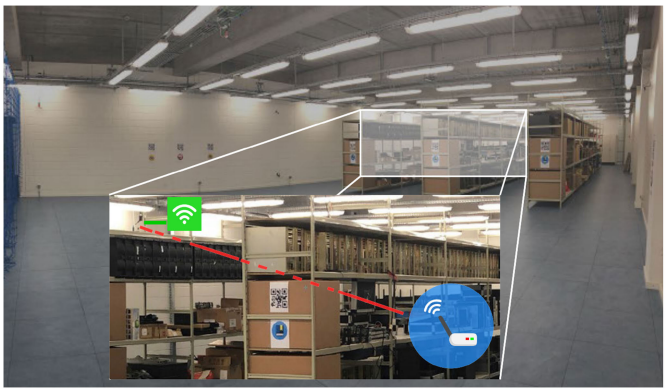


Fig. 4. Industrial environment of the IULCD dataset [33], [34].

based on a NVIDIA GPU 4060 8GB device, and the inference time tests are based on Intel Core CPU i5-13500H.

To evaluate the effectiveness of auxiliary features for NLOS classification, a baseline Mamba classifier using only pure CIR signals was also implemented. As discussed in Section IV, the Mamba architecture exhibits diminishing returns with increasing depth. Therefore, this article explored the optimal balance between resource consumption and performance for LightMamba by adjusting the number of Mamba layers and the encoding dimension. The specific hyperparameter settings used during the training process are detailed in Table II.

C. Evaluation Metrics

In classification tasks, common metrics for evaluating model performance include Accuracy, Precision, Recall, and $F1$-score. For the binary classification task of this article (NLOS versus LOS), NLOS is considered the positive class, and LOS is considered the negative class. The specific calculation formulae of these evaluation metrics are as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{11}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{14}$$

where true positive (TP) is the number of NLOS samples correctly predicted as NLOS, true negative (TN) is the number of LOS samples correctly predicted as LOS, false positive (FP) is the number of LOS samples incorrectly predicted as NLOS, and false negative (FN) is the number of NLOS samples incorrectly predicted as LOS.

TABLE II
HYPERPARAMETER SETTINGS

Hyperparameter	Value
Batch Size	128
Epochs	300
Learning Rate	0.001
Weight Decay	0.0001
Loss Function	BCEWithLogitsLoss
Dropout Rate	0.3
Optimizer	AdamW
Patience	30

VI. RESULTS AND ANALYSIS

A. Effectiveness of Architecture Design and Optimal Balance Model

This section presents a comprehensive evaluation of the proposed LightMamba framework, focusing on validating its architecture design and identifying an optimal configuration that effectively balances high classification accuracy with the stringent resource constraints typical of edge deployments.

1) Effectiveness of Auxiliary Features: Table III lists the performance of LightMamba with/without auxiliary features and with varying encoding dimensions and layer numbers using the eWINE dataset. It clearly demonstrates the significant benefit of incorporating auxiliary features. Models using both CIR and auxiliary features consistently outperform models using only CIR data across all configurations and metrics. Specifically, for the configuration with a 32-D encoding and 1 layer (32-dim/1-layer for short), auxiliary features boost the $F1$-score substantially from 87.04% to 92.26%. This performance enhancement is achieved with only a marginal increase in parameter count; for the 32-dim/1-layer configuration, parameters increase slightly from 25.4k to 25.8k.

Notably, a consistent reduction in inference time was observed across all configurations when auxiliary features were included. We speculate that the rich environmental context provided by these features acts synergistically with Mamba's selective mechanism. This contextual "prompt" could potentially guide the model to more rapidly identify salient CIR segments, thereby accelerating inference convergence.

2) Diminishing Returns and Optimal Configuration: Further analysis of the results using both CIR and auxiliary features in Table III reveals a clear pattern of diminishing returns with increasing model complexity, primarily defined by coding dimension and layer number. While scaling from the 16-dim/1-layer to the 32-dim/1-layer configuration yields a significant $F1$-score gain of 1.15%, this comes with a

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TABLE III
PERFORMANCE COMPARISONS OF LIGHTMAMBA WITH/WITHOUT AUXILIARY FEATURES AND WITH VARYING ENCODING DIMENSIONS AND LAYER NUMBERS FOR THE EWINE DATASET

Type	Coding dim	Num layers	Accuracy	Precision	Recall	F1-score	Parameters	Inference time (ms)
Only CIR	16	1	87.57%	89.89%	84.67%	87.2%	**10641**	2.353
	32	1	86.98%	86.61%	87.48%	87.04%	25338	2.621
	64	2	87.1%	88.58%	85.05%	86.83%	117953	5.745
	128	4	86.83%	87.31%	86.19%	86.75%	659329	14.635
CIR and auxiliary features	16	1	91.21%	91.57%	90.65%	91.11%	10913	**1.431**
	32	1	92.38%	**93.16%**	91.37%	92.26%	25793	1.610
	64	2	92.6%	92.36%	92.76%	92.56%	119041	3.101
	128	4	**93.1%**	93.13%	**92.95%**	**93.04%**	661505	7.466

TABLE IV
ABLATION ANALYSIS OF LIGHTMAMBA'S ARCHITECTURAL COMPONENTS CONDUCTED ON THE EWINE DATASET

Configuration	Accuracy	Precision	Recall	F1-score	Parameters	Inference time (ms)
Full CIR, Addition	91.81%	92.16%	**91.71%**	91.71%	54241	**1.226**
T=4, Addition	92.17%	92.58%	91.56%	92.07%	29857	1.506
T=8, Addition	**92.38**%	**93.16**%	91.37%	**92.26**%	25793	1.610
T=127, Addition	90.17%	92.7%	87.06%	89.79%	**21985**	7.063
T=8, Concatenation	91.9%	92.09%	91.56%	91.83%	25697	1.646

moderate cost increase, adding 14.9k parameters and 0.39 ms inference time. In contrast, further expansions to the 64-dim/2-layer and 128-dim/4-layer configurations offer only marginal $F1$-score gains of 0.30% and 0.48%, respectively. These minimal gains are achieved at the cost of substantial increases in parameters, which are approximately 4.6 and 25.6 times more than the 32-dim/1-layer configuration, and correspondingly longer inference times, approximately 1.5 and 3.7 times longer.

This observation aligns with the expected behavior of SSMs and underscores the critical need to balance performance against resource constraints for practical edge deployment. Based on this trade-off analysis, the LightMamba configuration employing a 32-D encoding, a single Mamba layer, and using both CIR and auxiliary features is recognized to achieve the optimal balance. This 32-dim/1-layer "all features" configuration achieves a high $F1$-score of 92.26% while maintaining high resource efficiency, requiring only 25.8k parameters and a rapid inference time of approximately 2.0 ms. This makes it well-suited for resource-constrained applications. Consequently, this optimally balanced configuration is used for subsequent comparisons against baseline and SOTA methods.

3) Ablation Analysis of Architectural Components: Table IV presents a detailed ablation study conducted on the eWINE dataset to meticulously assess the impact of LightMamba's primary architectural components. The investigation first examined the role of CIR segmentation by comparing a model processing the entire, unsegmented CIR sequence against those employing segmentation. The results clearly indicate the benefits of segmentation; for instance, using eight segments ($\boldsymbol{T} = 8$) improved the $F1$-score to 92.26% from 91.71% achieved with the full CIR, while simultaneously reducing the parameter number by over 52%, from 54.2k to 25.8k. This highlights segmentation's effectiveness in enhancing both performance and computational efficiency by allowing the model to focus on relevant local temporal dynamics.

Furthermore, the sensitivity to the number of segments $\boldsymbol{T}$ was explored. Increasing the segmentation granularity $\boldsymbol{T}$ from 4 to 8 led to an improvement in $F1$-score and a reduction in parameters. However, employing excessive segmentation, such as $\boldsymbol{T} = 127$, resulted in a pronounced degradation in $F1$-score to 89.79% and a significant increase in inference time, despite requiring the smallest parameter number. This demonstrates a clear trade-off issue, indicating that the optimal number

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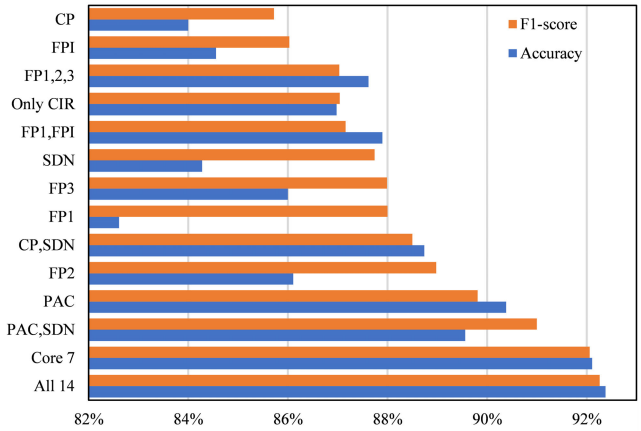


Fig. 5. Performance of the LightMamba model using a single and different combinations of auxiliary features with the eWINE dataset.

of segments exists, which captures sufficient temporal details without fragmenting the sequence excessively or incurring prohibitive computational overhead during sequence modeling. For the eWINE dataset, $\boldsymbol{T} = 8$ can be regarded as the optimal choice under these considerations.

Finally, the study evaluated the proposed dual-branch feature fusion method, which uses element-wise addition, against a common alternative using simple feature concatenation. With the optimal segment number $\boldsymbol{T} = 8$, the addition-based fusion achieved a superior $F1$-score of 92.26% compared to 91.83% obtained with concatenation, while maintaining a comparable parameter number and inference time. This result validates the effectiveness of the element-wise addition strategy within the dual-branch structure. Collectively, these ablation results confirm the rationale behind LightMamba's architecture and pinpoint the configuration of the optimal balance between accuracy and efficiency to be used in subsequent comparisons.

4) Contribution Analysis of Different Auxiliary Features: All three datasets contain seven channel features reflecting the signal propagation characteristics, which include CIR power (CP), first path index (FPI), amplitudes of the three strongest paths (FP1, FP2, FP3), standard deviation of noise (SDN), and preamble accumulation count (PAC). The eWINE dataset contains seven additional configuration features describing hardware-specific settings.

To further understand the role of these auxiliary features, we conducted an ablation study to analyze their individual and combined contributions by taking the eWINE dataset as an example. A single or different combinations of auxiliary features were evaluated, with the results shown in Fig. 5. All the training processes used the same optimal hyperparameter configuration described in Section VI-A2.

A key finding from Fig. 5 is the contrary effects of redundant and complementary features. For instance, combining highly correlated features that describe path amplitudes, namely FP1, FP2, and FP3, results in an $F1$-score of 87.03%, which is lower than using any single one of these features, with the highest $F1$-score achieved by using FP2 alone, reaching 88.98%. In contrast, fusing orthogonal information from PAC and SDN, which describe signal quality and channel background conditions, respectively, boosts the $F1$-score to 91.00%. This corresponds to a significant improvement over using either PAC or SDN alone, with $F1$-scores of 89.81% and 87.74%, respectively. The observed contrary effects highlight that performance gains stem from the effective fusion of complementary physical indicators.

Compared to using the raw CIR data without any auxiliary features with a baseline $F1$-score of 87.04%, the inclusion of auxiliary information generally leads to a substantial performance improvement. Notably, the combination with only the seven core channel features (Core 7) achieves an $F1$-score of 92.06%, which is nearly identical to the $F1$-score of 92.26% achieved with all 14 available features (All 14). This indicates that the additional hardware-specific parameters are largely redundant for this task. It also confirms that the model's high performance is derived from learning physically meaningful channel characteristics, not from overfitting to specific device configurations. This inherent focus on physically meaningful features is fundamental to the model's robust cross-dataset performance and high transferability.

B. Performance Comparison Across Different Datasets

To evaluate the effectiveness of the proposed LightMamba framework, we compared its performance against several well-established baseline models, including CNN, LSTM, RNN, and transformer. The comparison was conducted across the OIUD, eWINE, and IULCD datasets, using the optimally balanced LightMamba configuration obtained above. Table V presents the detailed results.

Across all three datasets, LightMamba consistently demonstrates superior performance compared to the baseline models. On the OIUD dataset, LightMamba achieves an exceptional $F1$-score of 95.20%, significantly surpassing the second-best baseline model, RNN, with an $F1$-score of 83.42%. Similarly, on the IULCD dataset, LightMamba attains an $F1$-score of 91.45%, markedly higher than the baseline models scoring 85.08% with RNN and 86.21% with transformer. For the eWINE dataset, LightMamba leads with an $F1$-score of 92.26%, outperforming all baseline models, including the transformer scoring 91.47%.

In addition to classification accuracy, another key advantage of LightMamba lies in its remarkable resource efficiency. As shown in Table V, LightMamba achieves its SOTA performance with drastically smaller parameter numbers compared to all baseline methods. For the OIUD dataset, LightMamba requires only approximately 22k parameters. This represents a reduction of over 50% compared to CNN and is substantially fewer than RNN, transformer, and especially LSTM, constituting a parameter reduction of about 96% relative to LSTM. This parameter reduction holds for the other two datasets as well. For eWINE and IULCD, LightMamba uses merely 25.8k and 26.1k parameters, respectively, while LSTM requires over 2 million parameters, and the other baseline models generally require 200k~700k parameters.

The comparative analysis clearly demonstrates LightMamba's superiority. LightMamba achieves higher NLOS identification accuracies across diverse datasets with significantly higher computational efficiency, requiring orders

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TABLE V
PERFORMANCE COMPARISONS OF LIGHTMAMBA WITH BASELINE MODELS

Dataset	Model	Accuracy	Precision	Recall	F1-score	Parameters
	CNN	82.62%	78.54%	81.23%	79.86%	47962
	LSTM	81.36%	81.21%	83.23%	82.21%	545064
OIUD	RNN	84.71%	86.39%	80.64%	83.42%	143122
	Transformer	83.67%	84.58%	79.93%	82.19%	160697
	LightMamba	**93.22%**	**97.67%**	**92.86%**	**95.20%**	**21864**
	CNN	90.29%	91.29%	88.93%	90.09%	206849
	LSTM	89.64%	90.21%	88.78%	89.49%	2173185
eWINE	RNN	89.83%	91.45%	87.73%	89.55%	592641
	Transformer	91.64%	92.80%	90.17%	91.47%	700417
	LightMamba	**92.38%**	**93.16%**	**91.37%**	**92.26%**	**25793**
	CNN	90.36%	84.57%	81.37%	82.94%	213154
	LSTM	87.97%	84.71%	84.56%	84.63%	1251721
IULCD	RNN	86.45%	84.26%	85.91%	85.08%	282845
	Transformer	92.47%	90.10%	82.64%	86.21%	311809
	LightMamba	**94.45%**	**93.10%**	**89.86%**	**91.45%**	**26137**

of magnitude fewer parameters than conventional deep-learning sequence models. This unique combination of high accuracy and extreme lightness makes LightMamba a highly effective and practical solution for resource-constrained UWB NLOS identification applications.

The performance variations of LightMamba across different datasets can be attributed to their different channel property and environmental complexity. The highest $F1$-score of 95.20% is achieved with the OIUD dataset, which we attribute to the dataset's shorter CIR sequence of only 150 points. In a shorter sequence, the key discriminative features of NLOS propagation, such as the transmission delay of the first path, are more salient and less likely to be obscured by signal interference or noise. For the eWINE and IULCD datasets with a much longer CIR sequence of 1016 points, $F1$-scores up to 92.26% and 91.45% can still be achieved. This validates LightMamba's ability to effectively model long-range dependencies and capture sparse critical features even in extended noisy signals. The slightly lower $F1$-score for IULCD possibly reflects its complex industrial environment, where severe multipath effects pose a challenge even for the advanced method of LightMamba.

C. Comparison Between LightMamba and SOTA Models

We compared LightMamba with some SOTA methods, including FCN-Attention [10], TinyML [12], and 1D-CLANet [9]. The comparison was mainly based on the eWINE dataset, considering the lack of SOTA-model-derived results in previous studies for the OIUD and IULCD datasets. However, the performance with OIUD and IULCD using LightMamba was also given. The results are listed in Table VI.

LightMamba achieves consistently robust performance across different datasets, achieving high $F1$-scores of 95.20%, 92.26%, and 91.45% for the OIUD, eWINE, and IULCD datasets, respectively. It significantly outperforms FCN-Attention for the eWINE dataset. Such a performance underscores the model's effectiveness and generalization capability. We attribute this to the synergy between Mamba's inherent strength in modeling long-range dependencies within CIR sequences and the rich context provided by auxiliary features, which enables the effective identification of discriminative patterns.

The specialized 1D-CLANet model [9] achieves the highest $F1$-score of 96.89% for the eWINE dataset. This can be attributed to its architecture, which is highly optimized for feature extraction, a sophisticated combination of 1D-CNN, LSTM, and squeeze-and-excitation attention (SE-Attention) layers. However, such a model introduces 1.4 million training parameters as reported by Wang et al. [9]. Compared to 1D-CLANet, LightMamba contains just 25.8k parameters, corresponding to a reduction by over 54 times. This reflects the different design priorities of the two models. 1D-CLANet is optimized for peak accuracy at the cost a high computational burden. In contrast, LightMamba prioritizes efficiency and deployment feasibility for resource-constrained edge devices. By accepting a modest performance degradation, LightMamba achieves a dramatic reduction in model size, making it a more practical solution for real-world edge applications.

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TABLE VI
PERFORMANCE COMPARISONS OF LIGHTMAMBA WITH SOTA MODELS

Model	Dataset	Accuracy	Precision	Recall	F1-score	Parameters	Inference time (ms)
LightMamba	OIUD	93.22%	**97.67%**	**92.86%**	**95.20%**	**21864**	1.821
	eWINE	92.38%	93.16%	91.37%	92.26%	25793	**1.610**
	IULCD	**94.45%**	93.10%	89.86%	91.45%	26137	1.697
FCN-Attention [10]	eWINE	88.24%	85.85%	91.56%	88.62%	1097587	——
TinyML [12]	eWINE	93.1%	93.21%	93.13%	93.17%	——	1.690
1D-CLANet [9]	eWINE	**96.89%**	**96.92%**	**96.88%**	**96.89%**	~1400000	——
TL CIR-based [26]	OIUD	92.8%	——	——	——	454259	——

LightMamba's performance is comparable to that of TinyML for the eWINE dataset. TinyML's effectiveness stems from a highly customized feature engineering process that manually reduces the 1016-D CIR to a 50-D feature vector. While this dataset-specific processing can mitigate noise and improve precision, it also limits the model's direct applicability to new environments. Adapting TinyML to a dataset like OIUD, with an original 150-D CIR, would require substantial manual feature re-engineering. In contrast, LightMamba uses the complete CIR through its end-to-end learning approach. This design underpins its superior out-of-the-box generalization ability and deployment convenience across varied data structures and application environments.

The defining characteristic and principal advantage of LightMamba is its profound resource efficiency, critical for practical deployment. Requiring only 22k to 27k parameters represents a dramatic reduction, by approximately 20-fold compared to the TL CIR-based method [37] with roughly 454k parameters, and over 50-fold compared to FCN-Attention and 1D-CLANet with more than 1 million parameters. This significant parameter reduction makes LightMamba feasible to be deployed on memory-constrained edge devices like microcontrollers, where multimegabyte models are often impractical. Besides this static efficiency, LightMamba exhibits a dynamic efficiency with short inference time periods around 1.6–1.8 ms. Notably for eWINE, it is slightly faster than TinyML despite processing input sequences nearly 18 times longer, highlighting the computational efficiency of its core Mamba mechanism.

D. Cross-Dataset Evaluations of Generalization Ability

To verify the model's generalization ability more strictly, we conducted cross-dataset evaluation experiments, i.e., training on one dataset and testing on another. More specifically, the LightMamba model was trained on the eWINE dataset and tested on the OIUD and IULCD datasets separately via zero-shot inference. To ensure consistency in input dimensions and facilitate meaningful comparison, we extended the CIR dimension of the OIUD dataset from 150 to 1016 by filling empty points with zero values. Besides, only 7 auxiliary features that are common across the three datasets, as mentioned in Section VI-A4, are retained for training and testing. All model hyperparameters were kept consistent with the optimal configuration identified in Section VI-A. For comparative analysis, two baseline models, CNN and RNN, are also trained and tested with the same experimental setup. The results are given in Table VII.

TABLE VII
CROSS-DATASET EVALUATION RESULTS OF LIGHTMAMBA AND BASELINE MODELS

Model	Test Dataset	Accuracy	Precision	Recall	F1-score
CNN	OIUD	37.33%	70.00%	19.95%	31.06%
	IULCD	64.29%	88.35%	62.47%	73.38%
RNN	OIUD	57.41%	72.49%	64.12%	68.05%
	IULCD	74.34%	81.29%	87.14%	84.24%
Light-Mamba	OIUD	70.41%	70.65%	99.51%	82.36%
	IULCD	78.67%	79.09%	98.96%	87.92%

According to Table VII, the baseline models experienced an obvious performance degradation in this cross-dataset experiment. The $F1$-score of the CNN model dropped to 31.06% on the OIUD dataset, suggesting that its convolutional mechanism relies on the local temporal patterns of the CIR, and consequently, its effectiveness was significantly reduced when faced with partially zero-filled inputs. Although RNN performed better than CNN, its $F1$-score on the OIUD dataset was still only 68.05%. Compared to CNN and RNN, LightMamba demonstrated significantly higher robustness, achieving $F1$-scores of 82.36% and 87.92% on the OIUD and IULCD datasets, respectively. This superior performance can be attributed to the selective mechanism of Mamba and the dual-branch architecture. Different characteristics between the training and testing datasets cause a domain shift and make the learned CIR features unreliable, leading to the complete failure of the CNN. Unlike CNN, LightMamba adaptively works by reducing its reliance on CIR features and using the more stable and generalizable auxiliary features. These comparative results

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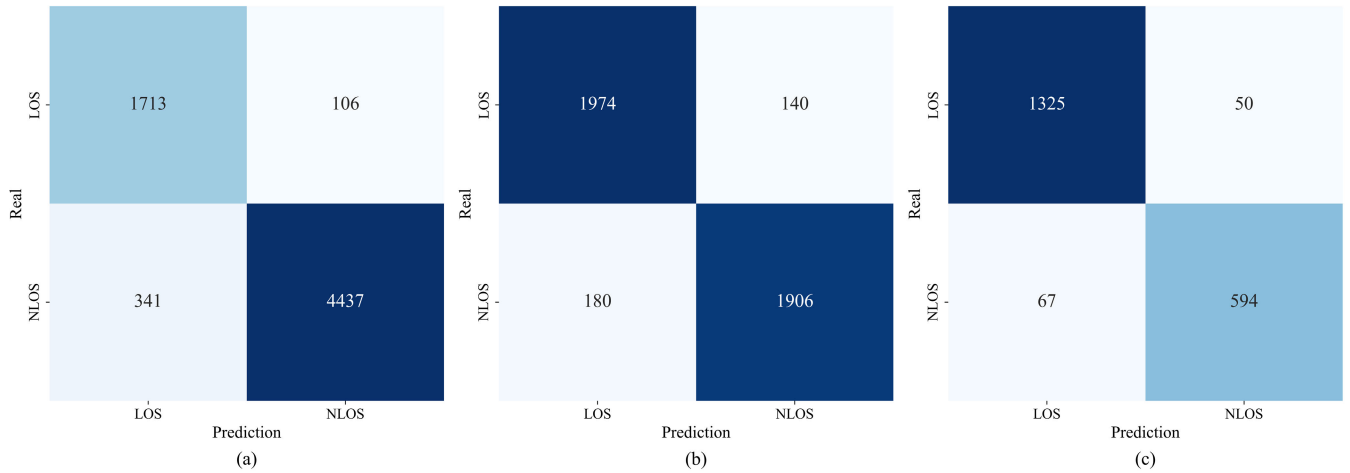


Fig. 6. Confusion matrices of LightMamba with different datasets. (a) OIUD. (b) eWINE. (c) IULCD.

indicate that our proposed method exhibits favorable cross-dataset generalization capability.

A noteworthy phenomenon is that LightMamba achieved high recall rates of approximately 99% on both test sets. This indicates that LightMamba tends to adopt a more sensitive classification strategy when facing feature unreliability. More precisely, this reflects a decision-making tendency that the model prefers to classify weaker LOS samples as NLOS to minimize the risk of misidentifying true NLOS samples as LOS and introducing potential positioning errors. Although this sensitive strategy led to a decrease in precision, we believe that this trade-off between precision and recall is beneficial for ensuring the reliability in industrial applications.

E. Analysis of Resource Consumption on an Edge Device

To demonstrate the feasibility of LightMamba for real industrial deployment, we deployed a quantized INT8 version of LightMamba on a specific edge device, i.e., the Arduino Nano nRF52840 MCU as shown in Fig. 2(b). Quantization is a common step for preparing neural networks for resource-constrained devices. TinyML was deployed in a similar way for comparison. Taking the eWINE dataset as an example, the resource consumptions of the unquantized FLOAT32 version and the quantized version are given in Table VIII.

The unquantized FLOAT32 version of LightMamba has a size of 113.61 KB and achieved an $F1$-score of 92.26%. After quantization, the model size was reduced to 36.37 KB. This quantization resulted in several changes to the resource consumption results. The peak RAM usage was reduced from 173.91 to 47.24 KB, well within the RAM memory of 256 KB available on the target device. Concurrently, the inference time was reduced from 6.357 to 2.824 ms. These improvements in resource efficiency were accompanied by only a 1.52% decrease in the $F1$-score, resulting in a final $F1$-score of 91.74% for the quantized version. These results suggest that the quantized version of LightMamba provides a lightweight memory usage and high computational efficiency while maintaining an acceptable classification performance. This further proves its feasibility for deployment on resource-constrained edge devices.

TABLE VIII
RESOURCE CONSUMPTIONS OF LIGHTMAMBA AND TINYML ON AN EDGE DEVICE WITH THE EWINE DATASET

Version	Model size (KB)	Accuracy	F1-score	Inference time (ms)	Peak RAM Usage (KB)
LightMamba (Unquantized)	113.61	92.38%	92.26%	6.357	173.91
LightMamba (Quantized)	36.37	92.03%	91.74%	2.824	47.24
TinyML (Unquantized)	21.9	93.14%	93.22%	1.69	——
TinyML (Quantized)	10.3	93.14%	93.18%	0.97	——

TinyML achieves faster inference and slightly higher $F1$-scores compared to LightMamba. This is largely attributed to its highly reduced input dimensionality of 64, which has to be accomplished through extensive feature engineering. In contrast, LightMamba processes the full 1030-D input. Although leading to a modest increase in inference time, this helps retain the significant advantages of an end-to-end approach, namely the superior generalization ability and elimination of manual dataset-specific engineering. This reflects the application-oriented design of LightMamba, which prioritizes out-of-the-box applicability and robustness in diverse environments.

F. Further Insights Into LightMamba's Performance

This section provides further insights into LightMamba's performance, as first visualized by the confusion matrices in Fig. 6. These matrices confirm the high classification fidelity across different datasets, characterized by strong diagonal dominance, indicating that correct classifications significantly outweigh misclassifications. The consistently small numbers of misclassifications underscore the model's reliability. The misclassification pattern of the IULCD dataset is slightly different from those of the other two datasets. To be specific, the proportion of true NLOS samples falsely identified as LOS ones is larger for IULCD, leading to a lower Recall of 89.86% as listed in Table VI. This likely reflects the severe multipath

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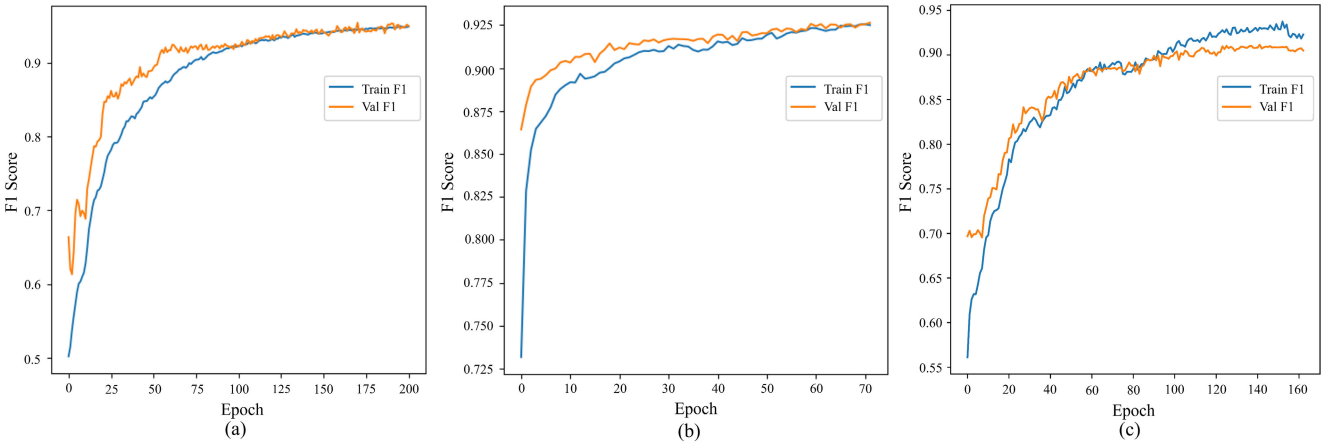


Fig. 7. Epoch-wise $F1$-score evolutions during training with LightMamba. (a) OIUD. (b) eWINE. (c) IULCD.

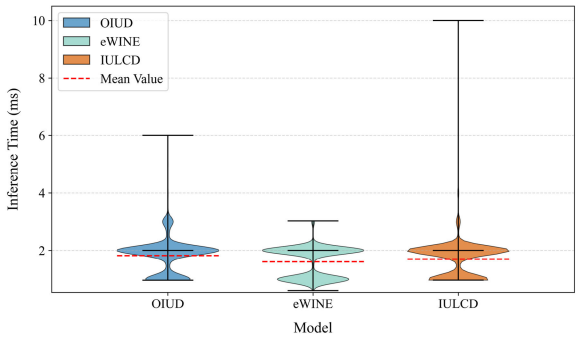


Fig. 8. Inference time distributions for the three datasets.

propagation in its industrial environment, highlighting the inherent challenge of modeling in such extreme conditions.

The epoch-wise $F1$-score evolutions during training with LightMamba for the three datasets are shown in Fig. 7. Curves in Fig. 7 clearly demonstrate effective learning and convergence across all datasets. Both training and validation $F1$-scores exhibit a rapid increase at the initial epochs, followed by a gradual stabilization at high performance levels, consistent with the high $F1$-scores reported above. The curve of validation $F1$-scores generally follows that of training $F1$-scores for the three datasets, particularly as training progresses. This indicates good model generalization performance and successful avoidance of significant overfitting.

These curves in Fig. 7 also reveal differences in convergence speed that appear to reflect dataset-specific complexity. The model achieves a near-peak $F1$-score very quickly for eWINE within 40 epochs, indicating its readily discriminated LOS and NLOS features. In contrast, the convergence on OIUD and IULCD is slower, implying a more complex feature distribution that requires more training iterations.

In addition to the reliable accuracy, the inference time distributions depicted in Fig. 8 visually substantiate LightMamba's exceptional computational efficiency, a crucial advantage over SOTA models. The violin plots consistently reveal low latency, with inference times clustered well below 2 ms for the majority of samples across the three datasets, aligning with the high average speeds discussed above. While the IULCD dataset shows slightly larger variability with an upper tail, processing time of most samples remains sufficiently short. This visual confirmation of consistently low latency, coupled with the drastically small parameter number discussed above, clearly demonstrates LightMamba's suitability for real-time and resource-constrained applications, offering a compelling blend of high performance and practical efficiency.

VII. Conclusion

This article proposes LightMamba, a lightweight UWB NLOS identification framework for industrial edge devices. It is elaborately designed by integrating selective SSMs with UWB signal characteristics. Comprehensive evaluations conducted across three challenging real-world UWB datasets, including OIUD, eWINE, and IULCD, validated LightMamba's superiority in both accuracy and efficiency. Specifically, LightMamba achieved an exceptional $F1$-score of 95.20% with the OIUD dataset with only 22k parameters. With the eWINE and IULCD datasets, it delivered high $F1$-scores of 92.26% and 91.45%, respectively, using only approximately 26k parameters. This performance represents a significant advancement over conventional deep-learning models. For instance, LightMamba reduces the parameter number by up to 96% compared to LSTM-based approaches while achieving superior $F1$-scores. Furthermore, LightMamba exhibits a rapid inference capability, consistently operating within 2 ms/sample, which is crucial for real-time industrial applications. These results demonstrate LightMamba's success to bridge the gap between high classification accuracy and stringent resource limitations typical of edge computing.

Future work could focus on enhancing LightMamba's peak performance by selectively integrating architectural elements inspired by 1D-CLANet, while also exploring advanced efficiency optimizations drawn from TinyML practices, such as aggressive quantization or feature space reduction. Furthermore, integrating LightMamba's reliable NLOS identification with complementary sensors, notably IMUs, presents a promising direction for developing high-precision, resource-constrained indoor localization systems based on multisensor fusion.

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following websites: https://github.com/JaronFontaine/Industrial-UWB-localization-CIR-dataset; https://github.com/JaronFontaine/UWB-dataset-from-an-office-industrial-and-university-environment; https://github.com/ewine-project/UWB-LOS-NLOS-Data-Set. Their code is available online: https://github.com/TeaWong1012/LightMamba.

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